Design Process Planning and Management

LAB EX-02: Dynamic analysis (Modal Analysis)

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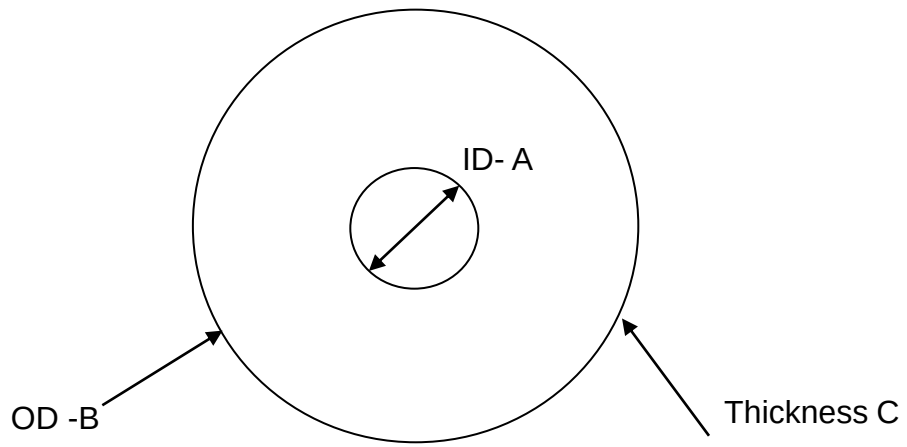
EX02: Dynamic Analysis

Target Date: 11.59 PM on 18th Jan

AIM
Perform a dynamics analysis to extract the first eight mode shapes and corresponding natural frequency, for the given geometry (circular plate with a central hole),
Tool: Fusion 360

Geometry:

Para meter	Details	Remarks
A	20 mm	ID
B	200 mm	OD
C	5 mm	Thickness



Material: Steel

F

Constraints: ID hole surface nodes are constrained in all three translational DOF

Material: MS

Procedure

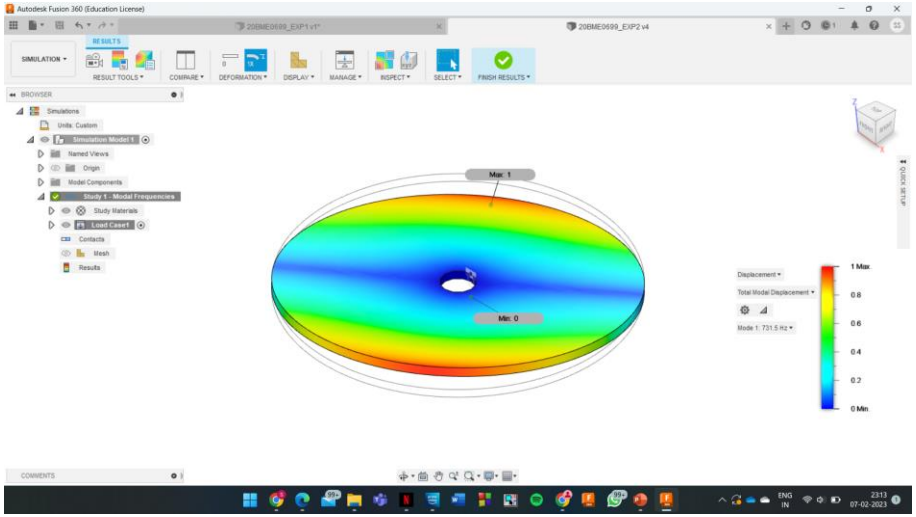
1. Create geometric model (Fusion 360) to the given dimensions in Fig.1
2. Save model EX02_ Roll No_GM
3. Go to simulation menu
4. Make analysis setup – Modal frequency
5. Assign material to the body
6. Apply structural constraints: All 3 DOF at the ID hole surface nodes
7. Define meshing parameter with the absolute size as **5 mm**
8 Generate mesh using linear tetrahedral elements
9 Check the model for input definition completeness
10 Solve the model
11 Post process the results and extract the first eight natural frequencies and capture the images of mode shapes clearly.
12 Animate all the mode shapes and identify the number of nodal diameter
13 Save the FE model EX02_ Roll No_FE
14 Repeate the analysis for absolute mesh size of 2.5 mm List the results the results in Table -1
15 Generate an automated report with a title : Modal Analysis
16 Complete the experiment report
17 Merge and Upload the auto REPORT GERNERATED BY fusion 360 and Experiment report, in your folder under EX02
18 Keep all the files safely to upload the same in VTOP when you receive communication later for upload

Table 1: Analysis results

Mode No	Mesh size: 5 mm		Mesh size: 2.5 mm		Remarks
	Frequency	Number of Nodal diameter	Frequency	Number of Nodal diameter	
1	731.5 Hz	1	524.9 Hz	1	
2	752.7 Hz	1	526.9 Hz	1	
3	2582.0 Hz	3	1909.0 Hz	3	

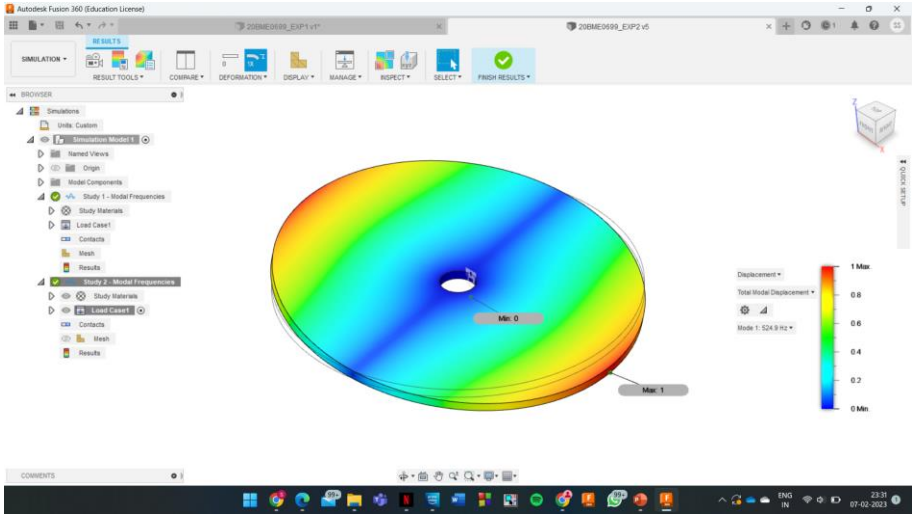
Fig. 1: Mode shape 1

5 mm



View 1

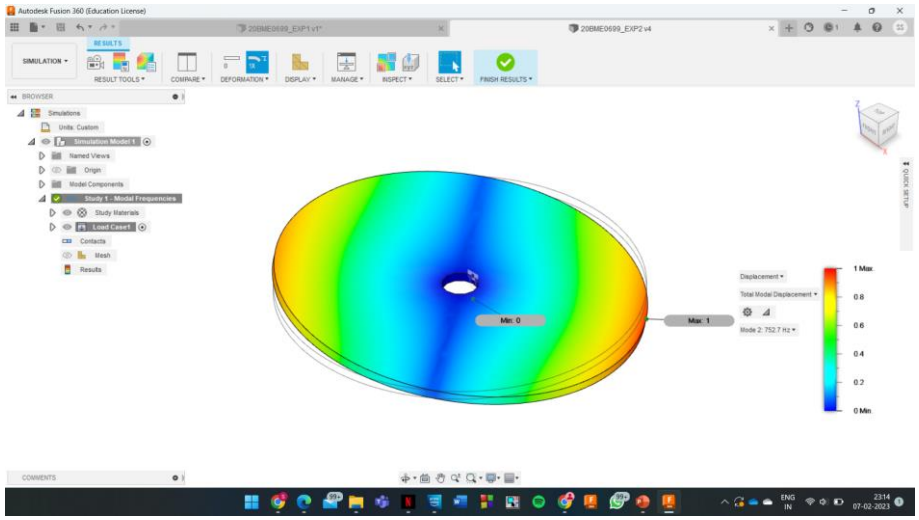
2.5 mm



View 2

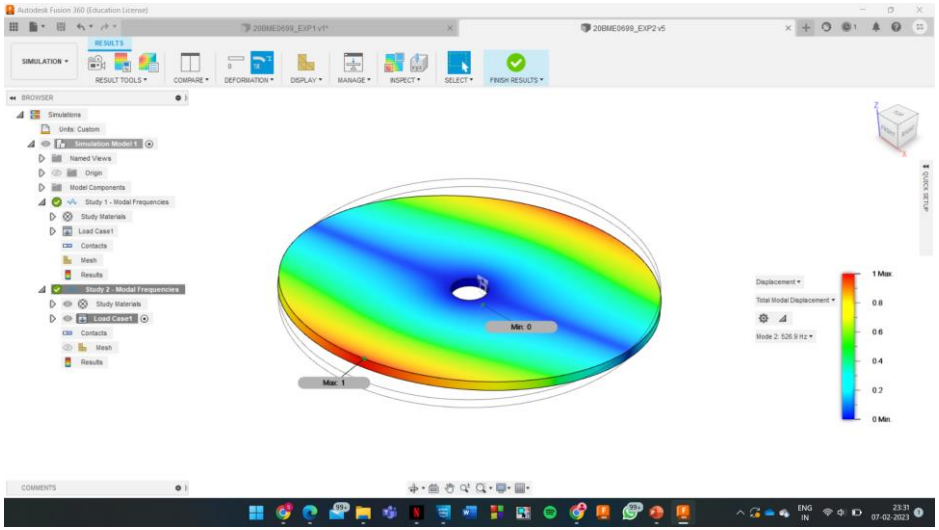
Fig. 2: Mode shape 2

5 mm



View 1

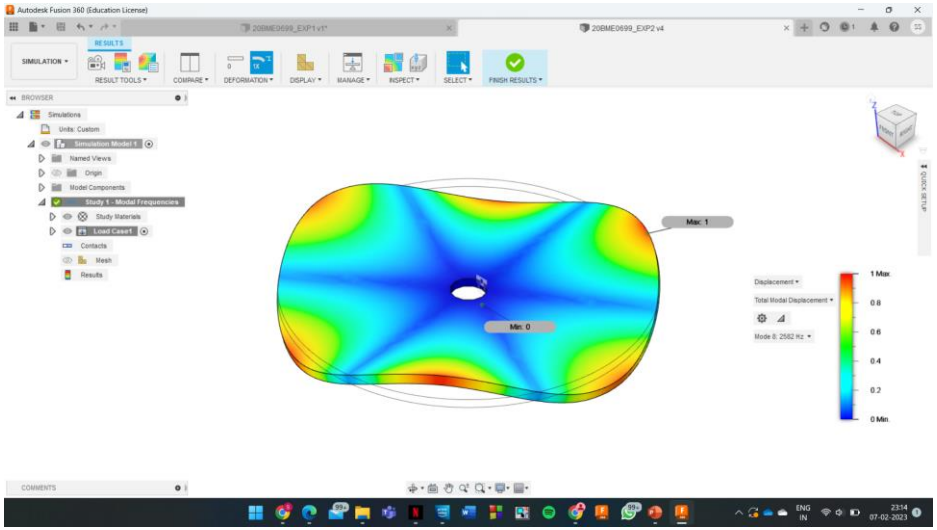
2.5 mm



View 2

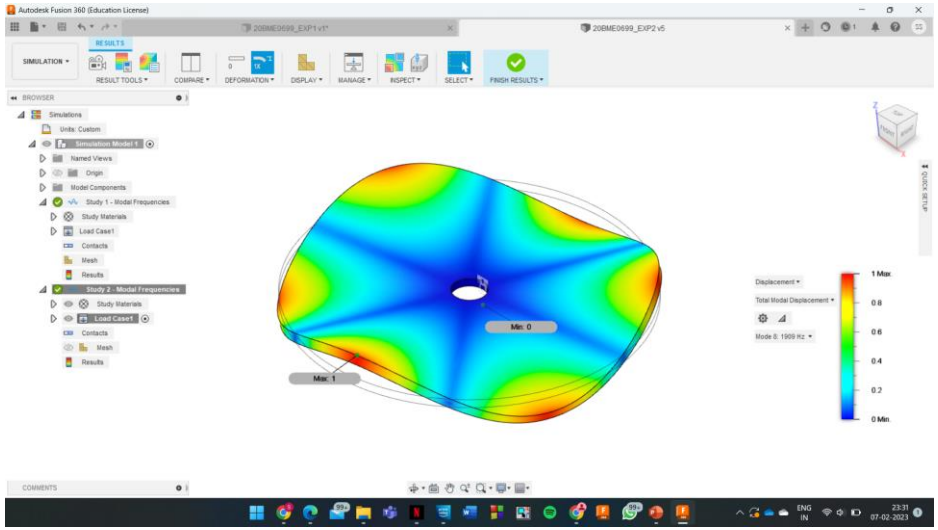
Fig. 8: Mode shape 8

5 mm



View 1

2.5 mm



View 2

Conclusion

The different frequency values were observed by using different mesh sizes of 2.5mm and 5mm respectively.
The frequency values increases from mode 1 to mode 8.

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SUJAL SRIVASTAVA